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# Position of the Lhasa terrane prior to India–Asia collision derived from palaeomagnetic inclinations of 53 Ma old dykes of the Linzhou Basin: constraints on the age of collision and post-collisional shortening within the Tibetan Plateau

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## SUMMARY

Seventeen sites were drilled from *ca.* 53 Ma old mafic dykes intruded in the Linzizong Formation of the Linzhou Basin for palaeomagnetic studies. From 10 sites a higher coercivity component demagnetized between 20 and 100 mT could be isolated. Detailed rockmagnetic analyses reveal Ti-rich titanomagnetite as the remanence carrier, which indicates that the rock is not much altered and the remanent magnetization is likely of primary origin. This is supported by a positive fold test. Tilt correction could be performed by the bedding of overlying fluvial lacustrine sediments and tilt angles of ignimbrite columns post-dating the dyke emplacement. Bedding corrected directions give an overall mean direction of D/I = 12.3°/27.2° ($\alpha_{95} = 10.6°$, $k = 21.7$, $N = 10$) corresponding to a palaeolatitude of 14.4° ± 5.8°N. Comparison with previous Cretaceous data mainly from the Takena Formation yields a stable position of the Lhasa terrane during Cretaceous and Early Eocene. The difference between expected palaeolatitudes determined from the APWP of Eurasia and observed ones reveals relative northward movement of the Lhasa terrane of *ca.* 1847 ± 763 km since early Eocene. This is attributed to indentation of India into Asia and implies a considerable amount of north–south crustal shortening. Together with the extent of 'Greater India', we can derive an age for the India–Asia collision between *ca.* 53–49 Ma with a 95 per cent confidence limit of ± 6 Ma.

**Key words:** Palaeomagnetism applied to tectonics; Rock and mineral magnetism; Continental margins: convergent.

## 1 INTRODUCTION

Collision between the Indian and Eurasian plates during the Cenozoic triggered the evolution of the Tibetan Plateau and the Himalayan mountain range. Prior to continental collision several mircoplates were accreted to southern Eurasia, of which the southernmost is the Lhasa terrane (Fig. 1). The Lhasa terrane formed the southern Eurasian margin since at least mid-Cretaceous. Palaeomagnetic data from the Lhasa terrane itself are essential to reconstruct the position of the pre-collisional southern margin of Eurasia. Several palaeomagnetic studies were carried out on the Late Cretaceous Lhunzhub Member of the Takena Formation of the Lhasa terrane (Pozzi *et al.* 1982; Westphal *et al.* 1983; Achache *et al.* 1984; Lin & Watts 1988; Tan *et al.* 2010). Only minor latitudinal motion of Eurasia since the beginning of Late Cretaceous is indicated by the Apparent Polar Wander Path (APWP, Besse & Courtillot 2002). Kapp *et al.* (2007) reported crustal shortening of minimum 160 km within the Lhasa terrane after deposition of the Takena Formation and prior to emplacement of the Linzizong Formation, but this limited deformation still allows using palaeomagnetic data of the Takena Formation to derive some rough constraint on the pre-collisional palaeolatitudinal position of the southern Eurasian margin. Palaeogene palaeomagnetic data from the Lhasa terrane are very limited (Westphal *et al.* 1983; Achache *et al.* 1984; Chen *et al.* 1993; Chen *et al.* 2010; Dupont-Nivet *et al.* 2010; Tan *et al.* 2010). In addition, reliability of part of the existing data is ambiguous because of a possible later overprint and partly insufficient sites for a good statistical evaluation (Westphal *et al.* 1983; Achache *et al.* 1984). Palaeogene palaeomagnetic results from the Lhasa terrane are particularly important as they can provide more direct constraints on the position of Eurasia's southern margin around the age of India–Asia collision (shortly before or after) than

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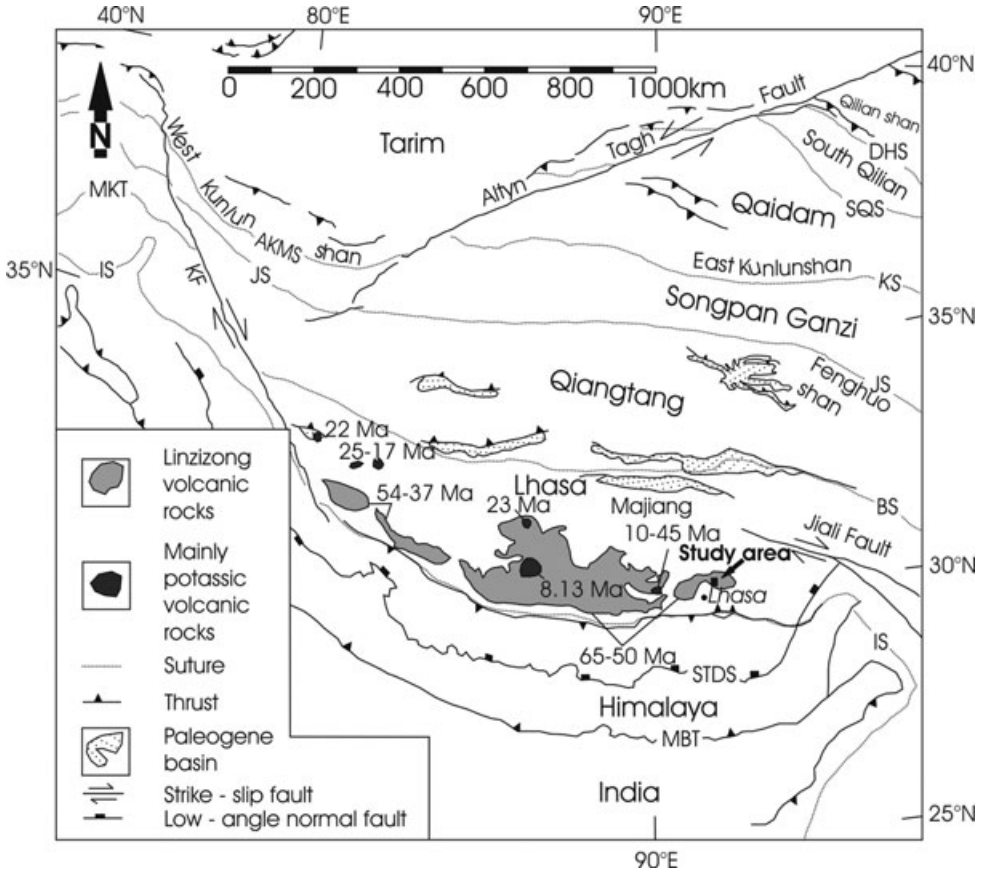



**Figure 1.** Map of the Tibetan Plateau showing the major terranes and spatial distribution of Linzizong volcanic rocks (modified after Ding *et al.* 2003). The study area is marked by a rectangle. Major faults: MBT: Main Boundary Thrust, STDS: Southern Tibetan detachment system. Main suture zones between the terranes: DHS: Danghe Nan Shan, SQS: Southern Qilian, KS: Kunlun, JS: Jinsha, BS: Bangong-Nujiang, IS: Indus-Yarlung.

Cretaceous ones. This study comprises palaeomagnetic investigations in the Lhasa terrane on Early Eocene mafic dykes intruding in the Linzizong Formation of the Linzhou area.

## 2 GEOLOGICAL SETTING

The Lhasa terrane is the southernmost of several crustal blocks, which were accreted to the southern margin of Eurasia prior to the Cenozic India–Asia collision (Fig. 1). It is separated from the Qiangtang terrane to the north by the Bangong-Nujiang suture and from the Indian affinity Tethyan Himalaya to the south by the Indus-Yarlung Suture Zone (IYSZ). The Bangong-Nujiang suture is regarded as the trace of a Mesotethyan ocean (Metcalfe 1988, 1996a,b), which opened during Permian or Triassic and closed in the Middle to Late Jurassic (e.g. Chang *et al.* 1986; Searle *et al.* 1987; Dewey *et al.* 1988; Metcalfe 2002). However, a later opening and closure of the Mesotethyan Ocean has been discussed recently. Baxter *et al.* (2009) found radiolarian faunas in the Lagkor Tso area, which indicate the presence of an oceanic basin until well into Early Cretaceous. This is supported by the occurrence of Early Cretaceous ophiolitic rocks at Nagqu (Chen *et al.* 2006). Prior to India–Asia collision the Lhasa terrane comprised an Andean-type margin (e.g. Coulon *et al.* 1986; Copeland *et al.* 1995). The Neo-Tethyan slab was subducted northwards along the southern margin of Asia. This previously low-angle subduction was probably followed by a slab rollback of the Neo-Tethyan oceanic lithosphere during Early Tertiary, which resulted in emplacement of the Linzizong Formation and development of the Gangdese magmatic arc in the southern Tibetan Plateau (Ding *et al.* 2003). The Linzizong Formation was emplaced between about 54 and 37 Ma in southwestern Tibet (Miller *et al.* 2000) and between about 69 and 43 Ma near Lhasa (e.g. Maluski *et al.* 1982; Coulon *et al.* 1986; Zhou *et al.* 2001, 2004; He *et al.* 2007). It is distributed extensively along the Gangdese Belt (Fig. 1). So far, detailed studies of the Linzizong Formation are confined to the Linzhou Basin (e.g. Dong 2002; Mo *et al.* 2003; Zhou *et al.* 2004; He *et al.* 2007; Mo *et al.* 2008; Chen *et al.* 2010; Dupont-Nivet *et al.* 2010; Tan *et al.* 2010), where it comprises a total thickness of *ca.* 3500 m (He *et al.* 2007). It can be separated into four units (Fig. 2; Yin *et al.* 1988; He *et al.* 2007). The lowest K-T unit consists of dacitic to rhyolitic tuffs, andesitic lava flows and ashflows. The overlying T1 unit is composed of a basal conglomerate interbedded with fine-grained arkosic sandstone and siltstone. Overlying rhyolitic ignimbrites are interbedded with lime-mudstone, siltstone, sandstone and volcanic breccias. The uppermost part of the T1 unit is provided by an ashflow. Within the T1 unit dyke swarms are common (He *et al.* 2007). The T2 unit comprises dacitic lapilli tuff. The uppermost T3 unit consists of interbedded sandstone, siltstone, mudstone and ashflows, with partly conglomerates at the base (He *et al.* 2007). The Linzizong Formation is gently folded and in general dips moderately to the north. It is unconformably underlain by strongly folded Cretaceous sedimentary sequences, which are in total over 3000 m thick (He *et al.* 2007). The lower part of the Cretaceous strata consists of pre-Albian fluvial and marginal marine deposits (Leeder *et al.* 1988; Leier 2005), which are conformably overlain by the Takena

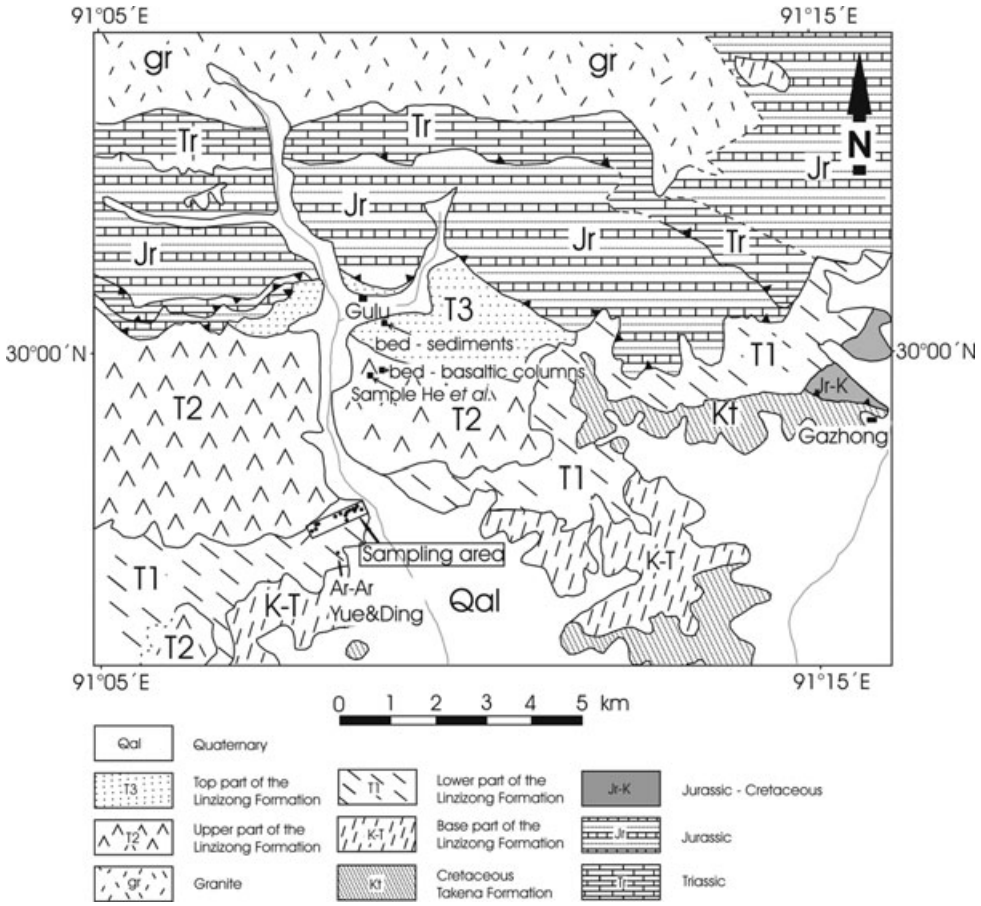



**Figure 2.** Geological map of the Linzhou area (modified after He *et al.* 2007) showing our site locations.

Formation. The Takena Formation can be separated into two members. The lower about 250 m thick Penbo Member comprises Aptian to Albian shallow marine limestones. The upper over 2000 m thick Lhunzhub Member consists of Upper Cretaceous fluvial redbeds (Yin *et al.* 1988). The maximum depositional age of the Lhunzhub Member is constrained to be younger than the conformably underlying Aptian-Albian Penbo Member. In addition, U-Pb zircon ages of fluvial sandstones cluster at 105 ± 2 Ma (Leier *et al.* 2004) supporting a maximum depositional age of Albian. The minimum depositional age is limited by the U-Pb ages of cross-cutting dykes, which cluster at *ca*. 52 Ma (He *et al.* 2007) and the unconformably overlying Linzizong Formation, whose youngest parts were dated to *ca.* 69 Ma by He *et al.* (2007). The continental character of the Lhunzhub Member indicates that closing of the Mesotethyan ocean was completed prior to deposition. It is interpreted as retro-arc foreland basin deposits (Leier *et al.* 2007). This study comprises palaeomagnetic investigations on a dyke swarm intruding into clastic, reddish sediments, which belong to the T1 unit of the Linzizong Formation (Fig. 3). Because the sedimentary rocks are covered by a strongly weathered surface layer and partly also by vegetation, no bedding attitudes could be determined. These basic dykes are exposed in the Linzhou area about 70 km NNE of Lhasa and about 1 km south of the village Nianbo. $^{40}Ar/^{39}Ar$ dating of one of the basaltic dykes (29°57.79′N/91°08.26′E) gave a plateau age of 52.9 ± 0.4 Ma (Yue & Ding 2006). The thickness of the dykes ranges from about 1.5 to 10 m. The ophitic texture and the relatively coarse grains reveal a subvolcanic character. There are three indications for the tectonic setting of the dykes. The first is provided by the bedding of fluvio-lacrustrine sediments (T3 unit) overlying the Linzizong Formation in Linzhou basin. The maximum age of the T3 unit is constraint to about 47 Ma, which is the age of the top of the T2 unit (He *et al.* 2007). The second structural control was derived from the tilting of ignimbritic columns (belonging to the T2 unit), close to the sampling site of He *et al.*, which yielded an age of about 47 Ma (Fig. 2); the columns are thus younger than the dykes. The third indication was obtained from the surface of a basaltic lava flow of the T2 unit. For tilt correction we used a mean value for the dip arrow determined by the first two methods, which give well-consistent results (azimuth/dip = 014°/26°, $\alpha_{95} = 4.1°$ and $N = 30$). Bedding of the lava flow is similar but due to the wavy surface the data show a large scatter and therefore were not considered for bedding correction.

## 3 SAMPLING AND ANALYTICAL METHODS

Cores from 17 sites were drilled using a portable rockdrill. Core orientation was done with a magnetic compass. Several tests showed that the indication of the compass was not influenced by the rocks and thus no sun-compass was required. The present day magnetic declination (IGRF2010, IAGA) at the sampling location is $D = 0.0°$. From the samples standard specimens of 2.5 cm diameter and 2.2–2.3 cm length were cut. Two pilot-specimens (twin specimens) of each site were progressively alternating field demagnetized and thermally demagnetized. Remanence directions were measured with

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**Figure 3.** Photograph of basic dykes intruding into reddish sediments of the T1 unit of the Linzizong Formation (seven of the dykes can be seen—marked by arrows).

a 2G Enterprises SQUID magnetometer 755R. For the alternating field demagnetization (AfD) a 2G6000 automatic degaussing system integrated in the SQUID magnetometer was used. Heating was done using a thermal specimen demagnetizer ASC scientific furnace (model TD-485 C). To detect possible changes in the magnetic mineralogy the bulk magnetic susceptibility was measured after each step of heating using a kappabridge KLY-2 (Agico). For AfD steps of 2–20 mT were chosen with a maximum field of 100 mT. Thermal demagnetization (ThD) was done using steps of 25–50 °C with a maximum temperature of 625 °C.

AfD was found to be more efficient for separation of the remanent magnetization components. Therefore the natural remanent magnetization (NRM) of eight to nine specimens per site was progressively alternating field demagnetized. To identify the ferro(i)magnetic minerals, one to four specimens per site were first alternating field demagnetized and afterwards applied to stepwise acquisition of isothermal remanent magnetization (IRM) using a pulse magnetometer MMPM9. Intensity of IRM was measured with a spinner magnetometer (MINISPIN-Molspin). Subsequently the saturation IRM (SIRM) was thermally demagnetized. High temperature (max. 700 °C) thermomagnetic runs were done from one powdered sample of each site using a CS-3 heating device coupled with kappabridge KLY-3 (Agico). Reflected light microscopy and domain structure observations using the Bitter method (Bitter 1932) were applied to 10 polished sections. Three polished sections were also used for energy dispersive X-ray (EDX) analysis in combination with scanning electron microscopy (SEM). An EDX-system OXFORD INCA Energy 200 Premium Si (Li) SATW-Detector and a SEM LEO Model 1450 VP were used. To reveal the relation between characteristic components and ferro(i)magnetic minerals, representative specimens were first applied to AfD with a maximum alternating field $\leq$40 mT to remove the lower coercivity component and afterwards treated by ThD.

## 4 MAGNETIC MINERALOGY

Identification of the ferro(i)magnetic minerals was performed in a detailed study including IRM acquisition, ThD of SIRM, high temperature thermomagnetic runs, reflected light microscopy and SEM/EDX analyses.

### 4.1 Reflected light microscopy and Bitter pattern observations

Reflected light microscopy reveals the occurrence of (titano) magnetite in all polished sections. However, shape of the crystals and alteration state vary considerably between different specimens. Grain sizes of (titano)magnetites show a high variation ranging

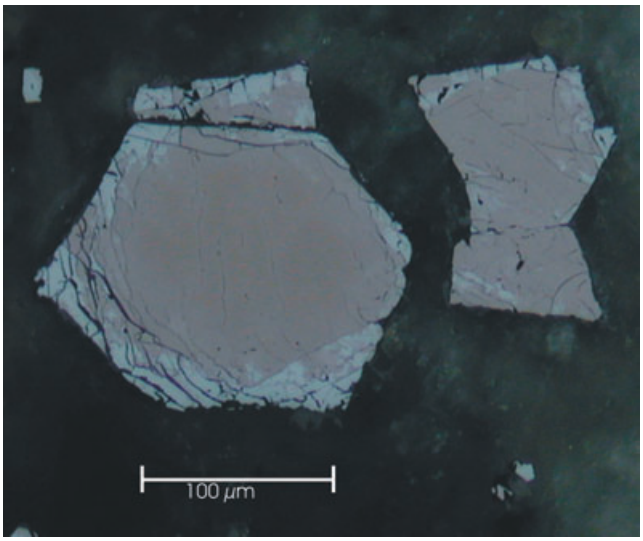



**Figure 4.** Partly maghemitized (titano)magnetite. Maghemitization is seen by the brighter colour and the shrinkage cracks around the margin. The hexagonal shape indicates a primary formation.

between 5 $\mu$m and *ca*. 500 $\mu$m. In all sections they were partly maghemitized (Fig. 4). In few cases further alteration to titanite, rutil and anatase was observed. The degree of alteration is higher along cracks and at the rim of the minerals. Some grains show shrinkage cracks due to maghemitization. Variation in alteration state between different sites is illustrated by the different frequency of cracks and the shape of the crystals. Smaller (titano)magnetite inclusions in silicates are less altered than minerals of the matrix. Partly present corona of small magnetite grains ($\leq$10 $\mu$m) around silicates display a secondary formation of magnetite due to exsolution or alteration of the silicates.

Domain observations using the Bitter colloid technique (Bitter 1932) were achieved to discriminate between Ti-rich titanomagnetite and magnetite. Final polishing of the sections was done with a suspension of amorphous silica microspheres (OP-U suspension, Struers) to remove the irregularly stressed surface layer produced by mechanical polishing (Hoffmann *et al.* 1987). Afterwards the stray fields at the intersection of domain walls with the polished surface were visualized by ester-based ferrofluid. Observed domain patterns, that is, wavy domain walls with varying intensities and zones without visible domain structures indicating uniaxial anisotropy and dominating stress control, are characteristic for Ti-rich titanomagnetites (Appel & Soffel 1984, 1985; Appel 1987; Halgedahl 1987) (Fig. 5). In magnetite straight domain walls at 180°, 71° and 109° angles are common (Soffel *et al.* 1982), however, volume domains would be only visible on planes oriented within a few degrees in respect to {110} (Kneller 1962).

To ensure that the observed features are domain images and not just structurally induced colloid accumulations, we applied external magnetic fields to check domain wall displacements. Applied fields changed the observed Bitter patterns and the resulting domain patterns after field removal were different from initial ones (Fig. 5).

## 4.2 SEM/EDX analyses

The chemistry of the Fe-Ti oxides was determined semi-quantitatively by EDX analysis. The results resemble a broad range from pure magnetite to Ti-rich titanomagnetite with compositions of $Fe_{2.6}Ti_{0.4}O_4$ to $Fe_{2.3}Ti_{0.7}O_4$. Compositions of $Fe_{2.4}Ti_{0.6}O_4$ (TM60)

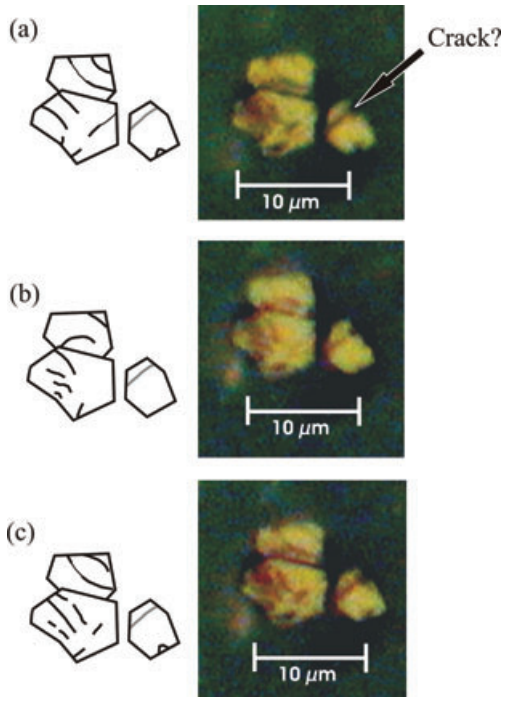



**Figure 5.** Photomicrographs of observed Bitter patterns with (a) zero external field, (b) applied magnetic field using a hand magnet and (c) zero external field after removing of the magnetic field. Sketches of the domain patterns are shown for clear illustration of the structures.

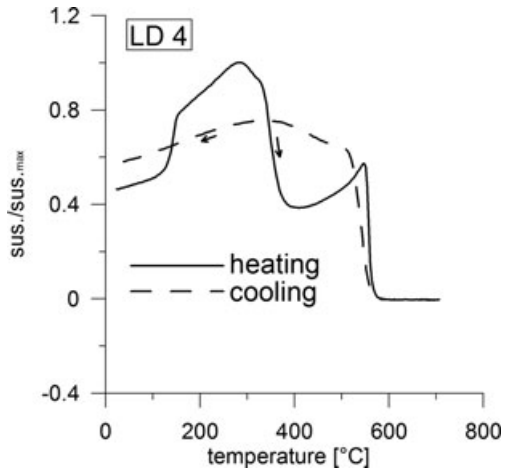



**Figure 6.** Magnetic susceptibility versus temperature heating (full line) and cooling (broken line) curves of a rock powder of site LD 4.

are typical for basic magmatic rocks. Some grains contain minor amounts of Al or Mg.

## 4.3 Rockmagnetic analysis

### *4.3.1 High temperature thermomagnetic runs*

Heating curves of high temperature thermomagnetic runs reveal that magnetic susceptibility is dominated by two distinct magnetic phases. A decrease in susceptibility between 500 and 580 °C indicates magnetite, which is present in all measured specimens. Most specimens show additionally a decrease between 300 and 350 °C (Fig. 6). Cooling curves reveal only magnetite, which indicates that

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the second phase is destroyed during heating. Possible explanations could be (i) inversion of maghemite to haematite or (ii) high-temperature oxidation of Ti-rich titanomagnetite with exsolution into magnetite- and ilmenite-near phases (Moskowitz 1981). As Ti-rich titanomagnetite is unambiguously identified by domain structure observations with a composition of $Fe_{2.6}Ti_{0.4}O_4$ to $Fe_{2.3}Ti_{0.7}O_4$ determined by EDX analyses the latter process is certainly dominant. Stoichiometric Ti-rich titanomagnetites around TM60 have a Curie temperature around 200 °C (O'Reilly 1984). The peak shape in the heating curve between 150 and 300 °C can be explained by low-temperature oxidation (LTO) during the heating process (in air), which increases the Curie temperature (Readman & O'Reilly 1972) before finally the titanomagnetites are destroyed during advanced heating.

#### *4.3.2 IRM acquisition*

IRM acquisition curves were analysed on basis of cumulative log-Gaussian (CLG) analysis using the software irmunmix2_2_1 and IRM_CLG1 (Kruiver *et al.* 2001). A coercivity parameter $B_{1/2}$ of the magnetic component is approximated by the applied field causing half of the SIRM to be acquired. For eight specimens IRM acquisition reveals saturation at fields $\leq$300 mT whereas six specimens were saturated between 300 and 500 mT (Fig. 7a); one specimen (LD 9–9–1) reached saturation only at a much higher field of *ca*. 1.3 T probably related to a higher content of haematite. A larger haematite content seems to be the exception as the S-ratio ($IRM_{-0,3T}$/SIRM) of all specimens except one is high (>0.9) supporting the dominance of ferromagnetic components. Because $M_s$ of magnetite is very high, its coercive force is dominated by the shape anisotropy giving a theoretical maximum for indefinitely long needles of about 300 mT. Saturation fields during IRM acquisition up to 500 mT is a further indication for the presence of Ti-rich titanomagnetite; phases around TM60 are dominated by stress anisotropy and irregular internal stresses may increase coercivity well above the limit of 300 mT for magnetite (Appel 1987).

CLG-analysis (Fig. 8) yields the presence of two components, a lower coercivity one ($B_{1/2}$ = 10–50 mT) probably related to magnetite and some contribution of multidomain titanomagnetite, and a harder one ($B_{1/2}$ = 80–120 mT), which is likely representing pseudo-single domain (or also single domain) Ti-rich titanomagnetite.

#### *4.3.3 ThD of SIRM*

ThD of SIRM shows a decrease in magnetization with unblocking temperatures corresponding to magnetite (max. 550 °C, dashed line example in Fig. 7b), and Ti-rich titanomagnetite (*ca.* 300 °C, full line example in Fig. 7b). The more continuous decrease reflects LTO of Ti-rich titanomagnetite. Because LTO occurs mainly at the crystal surface or in fissures, there are different oxidation stages within the grains leading to a broad range of Curie temperatures.

Ti-rich titanomagnetites are formed primary from rapidly cooling magmas. They are metastable and oxidize to titanomaghemite at ordinary temperatures (e.g. Özdemir 1987) or to several phases including magnetite and ilmenite at high temperatures (Grommé *et al.* 1969; Tucker & O'Reilly 1980). The direction of the remanent magnetization is not affected by maghemitization (Soffel 1972; Özdemir & Dunlop 1985). Therefore magnetic directions residing in Ti-rich titanomagnetite and/or titanomaghemite can be regarded as a primary magnetization. The strong abundance of Ti-rich titanomagnetite is a clear indication for a relatively fresh rock, because during advanced alteration processes these minerals would have been destroyed.

## 5 PALAEOMAGNETIC RESULTS

Initial NRM intensities of most sites are in the range of $10^2$–$10^3$ mA $m^{-1}$. Three sites (LD 2, LD 3 and LD 17) show two-orders higher NRM values up to 17.5 × $10^4$ mA $m^{-1}$ (Fig. 9). Separated magnetic directions of these sites are substantially different. However, magnetic susceptibilities at room temperature and SIRM values of all sites are of the same magnitude (Fig. 9) indicating a rather similar content of ferro(i)magnetic minerals. The only natural process causing such high NRMs is a lightning stroke. For that reason these sites were excluded from further analyses.

The degree of magnetic anisotropy (P) was determined for 13 sites by anisotropy of magnetic susceptibility (AMS) measurements. *p*-Values of all measured sites are very low ($p$ = 1.009–1.016). In some cases the maximum, intermediate and minimum axes of susceptibility within single sites show good grouping or are distributed on small circles. However, no correlation between different sites was found. Therefore possible minor influence of magnetic anisotropy on remanence directions is averaged out by calculating the mean of all sites and can be neglected.

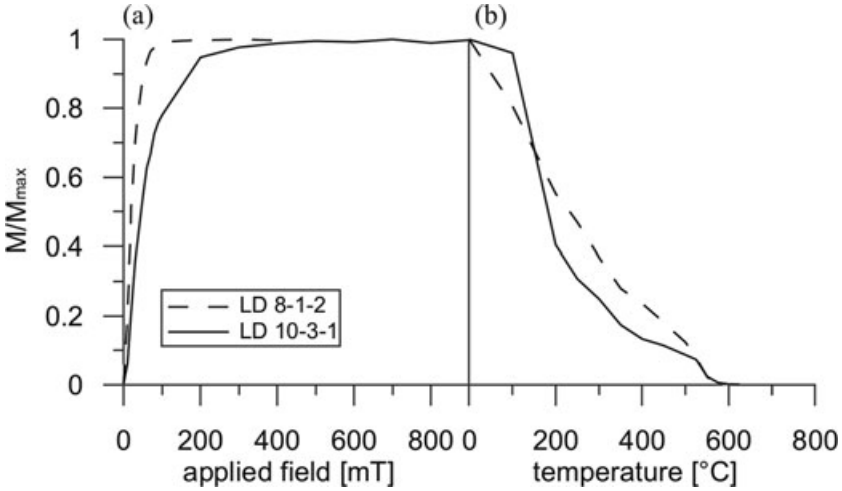



**Figure 7.** (a) IRM acquisition curves and (b) thermal demagnetization of SIRM of LD 8–1–2 and LD 10–3–1.

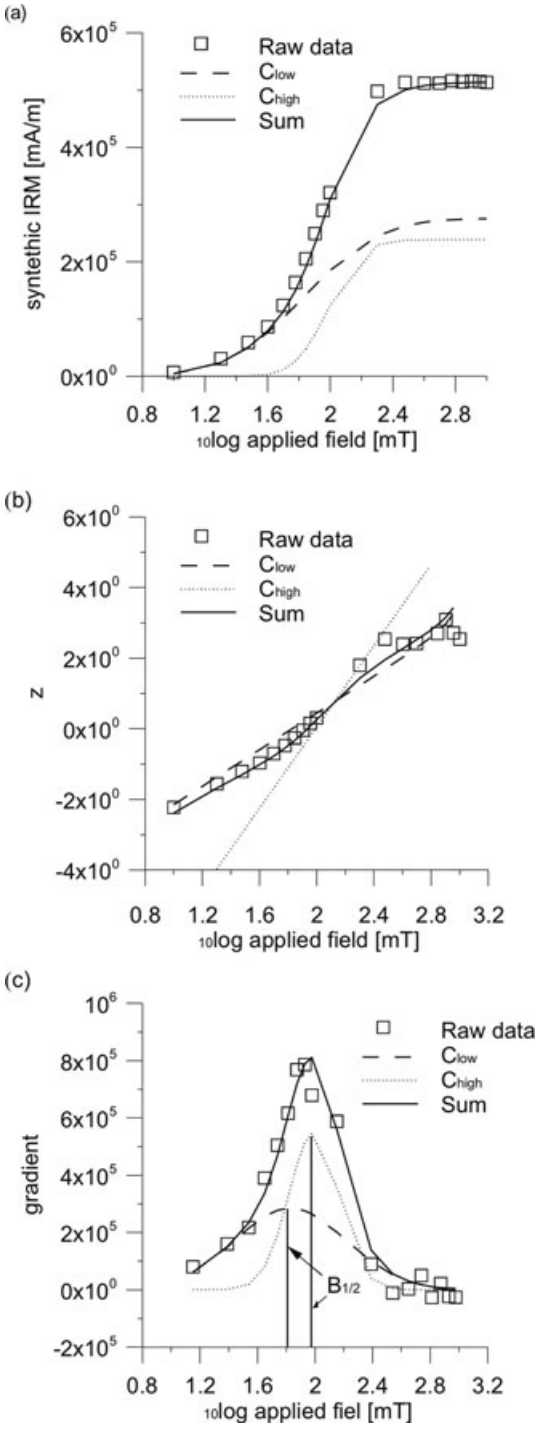



**Figure 8.** Results of cumulative log-Gaussian analysis of LD 5–1–2: (a) linear IRM acquisition plot, (b) gradient acquisition plot and (c) standardized acquisition plot; $C_{low}$: lower coercivity component, $C_{high}$: higher coercivity component, Sum: sum of components.

Magnetic directions were separated using principal component analysis. Eight to nine specimens per site were demagnetized by alternating fields with a maximum value of 100 mT. Variable components were observed in different samples. In most cases two components could be separated: a lower-coercivity component (LC-C), which was erased at maximum 25 mT, and a higher-coercivity component (HC-C), demagnetized at field intervals between 20 and 100 mT (Figs 10a and b). In some cases only a HC-C could be separated (Fig. 10c). Specimens of three sites comprise only the LC-C being almost completely demagnetized at 20 mT (Fig. 10d; Table 1). For statistical analyses, reverse polarity directions were inverted. With one exception directions residing in LC-C show normal polarity.

According to rockmagnetic analyses Ti-rich titanomagnetite and magnetite are the main ferro(i)magnetic minerals. AfD (max. applied field $\leq$45 mT) and subsequent ThD provide information about the relation between magnetic components and ferro(i)magnetic minerals: applying AfD the LC-C was completely demagnetized (Figs 11a and d); during subsequent ThD no considerable decrease of magnetization in the unblocking range of magnetite (above 400 °C) was revealed. Therefore the magnetite component was already erased by AfD and is related to the LC-C. In most cases one low temperature component (LT-C) unblocked in the range of Ti-rich titanomagnetite (150–300 °C) could be isolated by ThD. From some specimens also two or three antiparallel LT-Cs, which are all unblocked in the range on Ti-rich titanomagnetite could be separated (Figs 11b and c). These antiparallel components indicate that cooling of the dykes required a longer period of time, which includes one or two reversals of the Earth's magnetic field. Self-reversal of the titanomagnetites as reported by Krása *et al.* (2005) can be excluded, because reflected light microscopy clearly reveals single-phase titanomagnetites. Coincidence of remanence directions residing in LT-C (or inverted reverse polarity directions) and in the HC-C of a twin specimen treated by AfD up to 100 mT supports that both components have the same remanence carrier, that is, Ti-rich titanomagnetite.

Three sites comprise only a LC-C (sites LD 8, LD 12 and LD 13; Fig. 12a). From the remaining sites both an HC-C and an LC-C could be separated. However, the LC-C of most sites displays a high scatter (Fig. 12b). In contrast, with the exception of site LD 14, the HC-C of all sites shows good grouping with $k > 10$ (Van der Voo 1990; Figs 12b and c). Because of the uniform bedding no fold test could be applied exclusively on sites of the dykes. Therefore we combined nine sites of previous studies (Westphal *et al.* 1983; Achache *et al.* 1984) with the mean direction of the HC-C of all sites of this study. The Direction–Correction fold test after Enkin (2003) is positive within the 95 per cent confidence limit and yields an optimal degree of untilting at 88.7 per cent. The result of the parametric simulation fold test (Watson & Enkin 1993) reveals an analytical maximum at 88.4 per cent of untilting (Fig. 13). This indicates a primary character of the magnetic remanences. In addition, it is evident that the HC-C is related to Ti-rich titanomagnetite and therefore a primary origin is very likely. Ti-rich titanomagnetites can only be present when they formed from cooling magmas. They are metastable at normal temperatures and their high occurrence in the Linzhou dykes confirms that alteration of these rocks is minor. Intrasite occurrence of antipodal normal and reverse polarities also supports a primary nature. A reversal test including all HC-C site mean directions with $k > 10$ is positive with the classification 'indeterminate' after McFadden & McElhinny (1990) ($\gamma_c = 30.5$, $\gamma_{obs.} = 16{,}2$); the high $\gamma_c$ value is due to the relatively low number of sites (for the test sites were classified into normal and reverse groups depending on the pre-dominant polarity of the specimens). In Fig. 14 equal area projections of all bedding corrected site mean values with $k > 10$ are shown. The mean value of bedding corrected remanence directions residing in the LC-C yields D/I = 357.0°/39.7° ($\alpha_{95} = 16.5$, $k = 22.4$, $N = 5$). The LC-C is probably related to magnetite and a secondary magnetization is likely regarding the agreement of the LC-C mean direction in geographic coordinates (D/I = 349.2°/51.5°) with the present Earth magnetic

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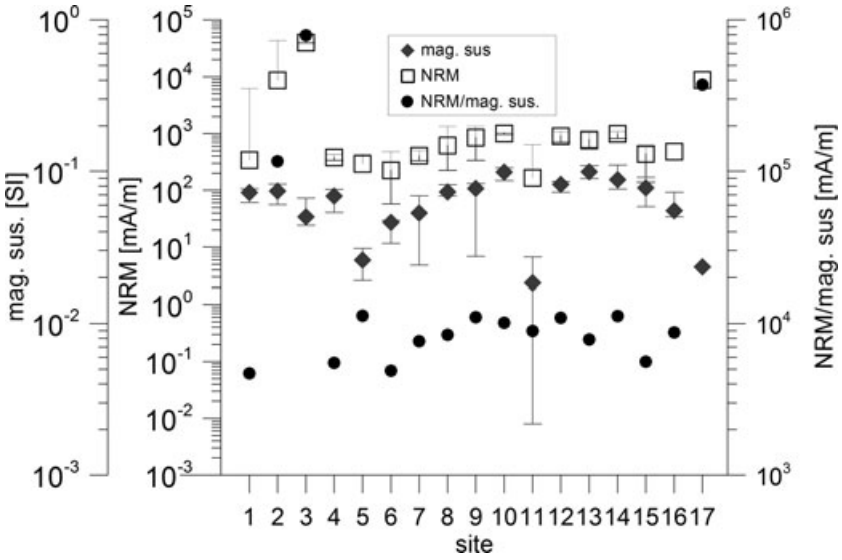



**Figure 9.** NRM values, magnetic susceptibilities at room temperature and the ratio of NRM to magnetic susceptibility, shown for each site (median values). For NRM values and magnetic susceptibilities also 25—and 75 per cent—-percentiles are displayed.

dipole field inclination (49.1°) at the Linzhou Basin (Fig. 14a). The equal area plot of site mean directions of the HC-C (Fig. 14b) shows good grouping with one possible outlier (LD 10). The 95 per cent confidence limit of the overall mean direction in geographic coordinates does not include the present Earth magnetic dipole field and therefore the HC-C is clearly a palaeoremanence. The overall bedding corrected mean direction including site LD 10 is D/I = 15.4°/27.2° ($\alpha_{95} = 9.7$, $k = 25.6$, $N = 10$); excluding this site we yield D/I = 13.4°/30.8° ($\alpha_{95} = 6.3°$, $k = 67.7$, $N = 9$). From these values palaeolatitudes of 14.4° ± 5.8°N and 16.6° ± 4.0°N result for the Linzhou area. Confidence limits are displayed by the d$p$ value of virtual poles calculated from the mean D/I. A possible explanation for the slightly differing remanence direction of LD 10 could be that the site reflects a transitional Earth's magnetic field. However, because rockmagnetic and palaeomagnetic characteristics of site LD 10 are similar to the other sites, we decided to consider the palaeolatitude including LD 10 for further discussion.

According to Van der Voo (1990) a minimum of 24 samples is necessary to average out palaeosecular variation (PSV). We therefore determined the mean remanence direction of the HC-C of all specimens of sites with $k > 10$. A mean value for the bedding corrected results of D/I = 16.4°/25.8° ($\alpha_{95} = 5.2°$, $k = 11{,}9$, $N = 68$) was achieved, which is similar to the results presented earlier; thus PSV was obviously already averaged out within sites.

## 6 DISCUSSION

Because of rockmagnetic analyses and results of stepwise unfolding we conclude that the HC-C has a primary origin. Thus the remanence age of the HC-C is approximately equal to the age of dyke emplacement, which was $Ar^{40}/Ar^{39}$ dated to *ca*. 53 Ma (Yue & Ding 2006). The direction residing in the HC-C partly shows reverse polarity (Fig. 10a). The occurrence of normal and reverse polarities within single sites indicates that remanence acquisition of different specimens was not contemporaneous, but occurred within a time span including at least one reversal of the Earth's magnetic field. A time-averaging effect can be also expected because the dykes were most likely formed in several magma pulses. Bons *et al.* (2004) describe a transport mechanism of magma, where the magma is transported in batches. This transport is stepwise and discontinuous in space and time and could be responsible for multiple intrusion events. Gudmundsson (1984) also reports on basaltic dykes formed in magma pulses. In an investigation on mafic dykes of Iceland, Gudmundsson (1995) reports that between injections in a multiple dyke times up to hundreds or thousands of years may elapse. The subvolcanic character of the dykes manifested in relatively coarse grains indicates a relatively slow cooling of the magma.

To test whether the PSV is really averaged out, we compared the precision parameter $k'$ calculated by the formula of McFadden (1980) with the precision parameter $k$ of our results. Using the latitude of our sampling sites (about 30°N) we revealed a $k'$-value of $k' = 31.7$. As our $k$-values of most sites as well as of the site mean directions and the mean of all specimens are below that value, PSV is obviously represented in the HC-C. The slightly higher scatter is probably due to some noise of remanence recording. In consideration of all these arguments we conclude that PSV is averaged out in the HC-C of our sampled dykes.

Bedding-corrected remanence directions residing in HC-C reveal a mean value of D/I = 15.4°/27.2°. The inclination yields a palaeolatitude of 14.4° ± 5.8°N of the Lhasa block for Early Eocene. Expected palaeodirections for the Linzhou area (30°N/91.1°E) at 53 Ma calculated from the APWP of Eurasia (Besse & Courtillot 2002) are D/I = 10.4°/50.2°, which corresponds to a palaeolatitude of 31 ± 2.9°N.

### 6.1 Comparison with previous studies

Fig. 15 displays expected palaeolatitude reference curves of India and Eurasia as well as palaeolatitudes calculated from present and previous palaeomagnetic investigations. Expected palaeolatitude reference curves were established from APWPs for stable India and Eurasia (Besse & Courtillot 2002) based on the present day geography with a reference point at the IYSZ (29°N/90°E). Previous palaeomagnetic investigations on rocks of the Early Eocene Linzizong Formation yield palaeolatitudes of 32.7 ± 4.8°N (Tan *et al.* 2010), 7.4 ± 4.6°N (T3 & T2 unit) and 10.2 ± 3.8°N (T1 & K/T unit) (Chen *et al.* 2010), 22.8 ± 3.3°N (Dupont-Nivet *et al.* 2010), 12.7° ± 6.3°N (Achache *et al.* 1984) and 8.3° ± 16.4°N (Westphal *et al.* 1983) at the reference point (29°N/90°E; see Table 2). Because the result of Westphal *et al.* is based upon

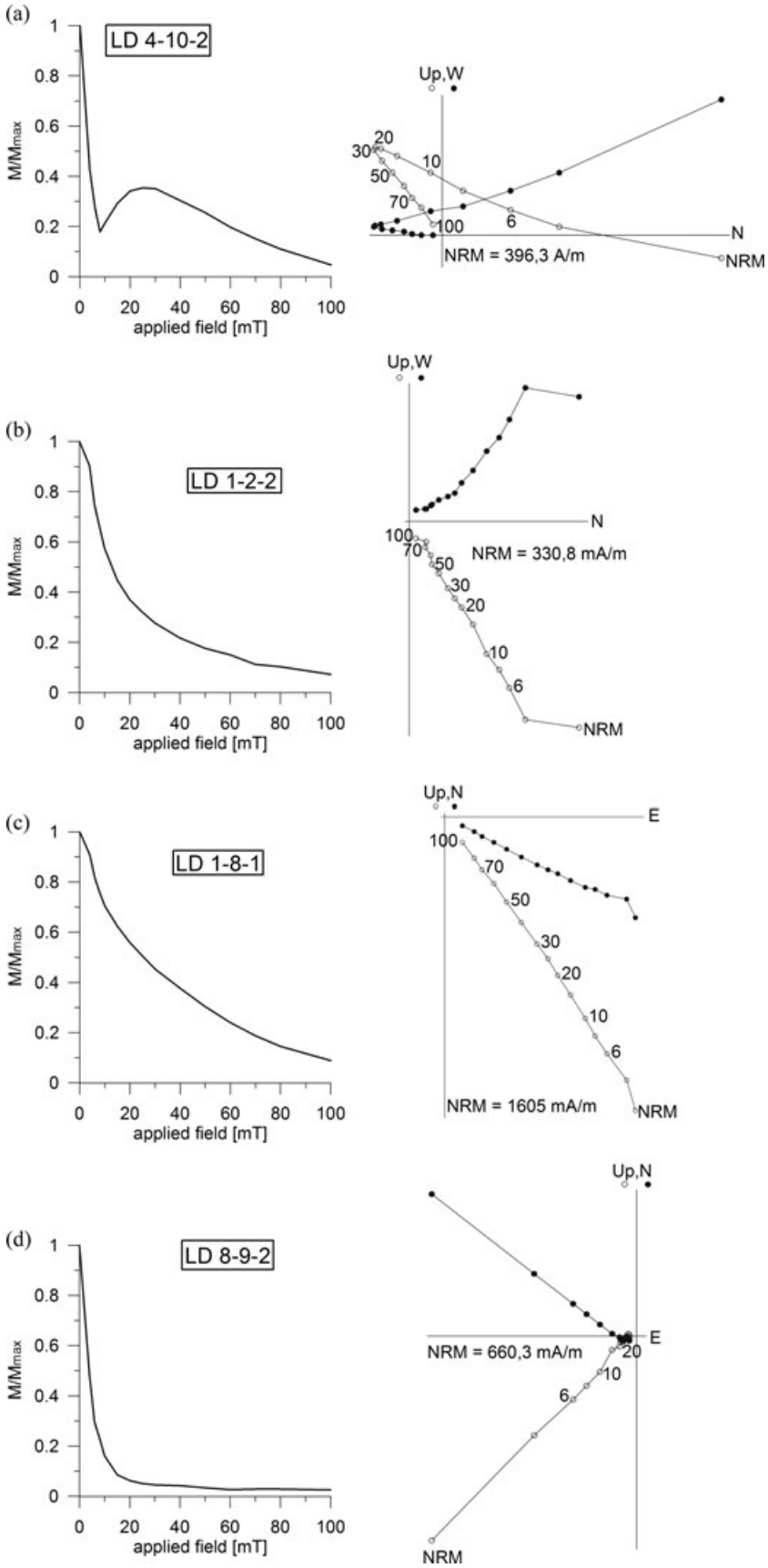



**Figure 10.** Results of alternating field demagnetization curves for representative specimens: intensity curves and Zjderveld diagrams (geographic coordinates) of (a) LD 4–10–2, (b) LD 1–2–2, (c) LD 1–8–1 and (d) LD 8–9–2.

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**Table 1.** Site mean values and statistical parameters of the palaeomagnetic results.

| | Site location | | | *In situ* | | Tilt corrected | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Site | Long. | Lat. | C | Dec. | Inc. | Dec. | Inc. | $k$ | $\alpha_{95}$ | $N_m$ | $N_{inc}$ | (norm./rev.) |
| LD 1 | 91°08,532′ | 29°58,408′ | HC-C | 12,7 | 54,5 | 15,8 | 40,7 | 15,1 | 24,4 | 8 | 4 | (3/1) |
| | | | LC-C | 313,7 | 51,0 | 327,9 | 45,0 | 23,6 | 19,3 | 8 | 4 | (4/0) |
| LD 4 | 91°08,389′ | 29°58,328′ | HC-C | 349,6 | 47,4 | 356,4 | 35,6 | 11,8 | 18,3 | 8 | 7 | (3/4) |
| | | | LC-C | 350,8 | 36,3 | 355,3 | 24,5 | 15,2 | 17,8 | 8 | 6 | (6/0) |
| LD 5 | 91°08,304′ | 29°58,314′ | HC-C | 9,6 | 37,3 | 11,8 | 23,8 | 48,5 | 11,1 | 8 | 5 | (1/4) |
| | | | LC-C | 10,3 | 38,4 | 12,5 | 24,9 | 3,2 | 44,7 | 8 | 6 | (6/0) |
| LD 6 | 91°08,231′ | 29°58,288′ | HC-C | 26,9 | 41,4 | 26,8 | 27,4 | 13,4 | 21,7 | 8 | 5 | (1/4) |
| | | | LC-C | 279,1 | 15,7 | 283,4 | 19,3 | 2,6 | 42,6 | 8 | 7 | (7/0) |
| LD 7 | 91°07,988′ | 29°58,204′ | HC-C | 9,1 | 36,2 | 11,2 | 22,7 | 16,6 | 12,2 | 10 | 10 | (6/4) |
| | | | LC-C | 354,8 | 51,7 | 1,5 | 39,3 | 13,8 | 73,4 | 10 | 2 | (2/0) |
| LD 8 | 91°07,960′ | 29°58,183′ | LC-C | 3,8 | 26,0 | 5,6 | 13,0 | 4,7 | 28,6 | 8 | 8 | (8/0) |
| LD 9 | 91°07,952′ | 29°58,122′ | HC-C | 16,6 | 38,5 | 17,9 | 24,6 | 19,8 | 10,5 | 11 | 11 | (6/5) |
| | | | LC-C | 0,0 | 40,8 | 3,9 | 28,0 | 15,6 | 20,0 | 11 | 5 | (5/0) |
| LD 10 | 91°07,903′ | 29°58,101′ | HC-C | 30,5 | 6,0 | 30,6 | −8,0 | 75,3 | 8,9 | 8 | 5 | (0/5) |
| | | | LC-C | 336,8 | 56,4 | 348,8 | 46,1 | 4,6 | 40,3 | 8 | 5 | (5/0) |
| LD 11 | 91°08,506′ | 29°58,391′ | HC-C | 14,6 | 40,4 | 16,3 | 26,6 | 144,8 | 4,6 | 8 | 8 | (0/8) |
| | | | LC-C | 19,4 | 59,6 | 21,2 | 45,7 | 3,0 | 46,6 | 8 | 6 | (6/0) |
| LD 12 | 91°08,455′ | 29°58,380′ | LC-C | 2,3 | 57,5 | 8,4 | 44,4 | 29,8 | 9,6 | 9 | 9 | (9/0) |
| LD 13 | 91°08,407′ | 29°58,340′ | LC-C | 0,2 | 64,9 | 8,6 | 51,8 | 20,7 | 13,6 | 8 | 7 | (7/0) |
| LD 14 | 91°08,396′ | 29°58,319′ | HC-C | 86,8 | 59,8 | 70,5 | 51,2 | 3,7 | 31,1 | 9 | 9 | (7/2) |
| | | | LC-C | 48,2 | 64,6 | 41,2 | 50,7 | 5,4 | 43,9 | 9 | 4 | (3/1) |
| LD 15 | 91°08,385′ | 29°58,307′ | HC-C | 2,8 | 52,4 | 7,9 | 39,4 | 78,0 | 6,9 | 9 | 7 | (4/3) |
| | | | LC-C | 323,8 | 57,0 | 339,0 | 48,8 | 6,4 | 39,6 | 9 | 4 | (4/0) |
| LD 16 | 91°08,092′ | 29°58,286′ | HC-C | 12,4 | 47,7 | 15,0 | 34,0 | 15,4 | 17,6 | 8 | 6 | (2/4) |
| | | | LC-C | 321,7 | 54,7 | 336,3 | 47,0 | 6,2 | 123,0 | 8 | 2 | (2/0) |
| mean HC-C including LD 10: | | | | 13,5 | 40,9 | 15,4 | 27,2 | 25,6 | 9,7 | | 10 | |
| mean HC-C excluding LD 10: | | | | 10,8 | 44,4 | 13,4 | 30,8 | 67,7 | 6,3 | | 9 | |
| mean LC-C: | | | | 350,2 | 51,5 | 357,8 | 39,6 | 27,5 | 13,6 | | 6 | |

Long.: longitude, Lat.: latitude, C: component, Dec.: declination, Inc.: inclination, $k$: precision parameter, $\alpha_{95}$: 95 per cent confidence angle, $N_m$: number of specimens measured, $N_{inc}$ (norm./inv.): number of specimens included in statistics (normal polarity/reverse polarity); only sites with $k > 10$ were included in the mean values.

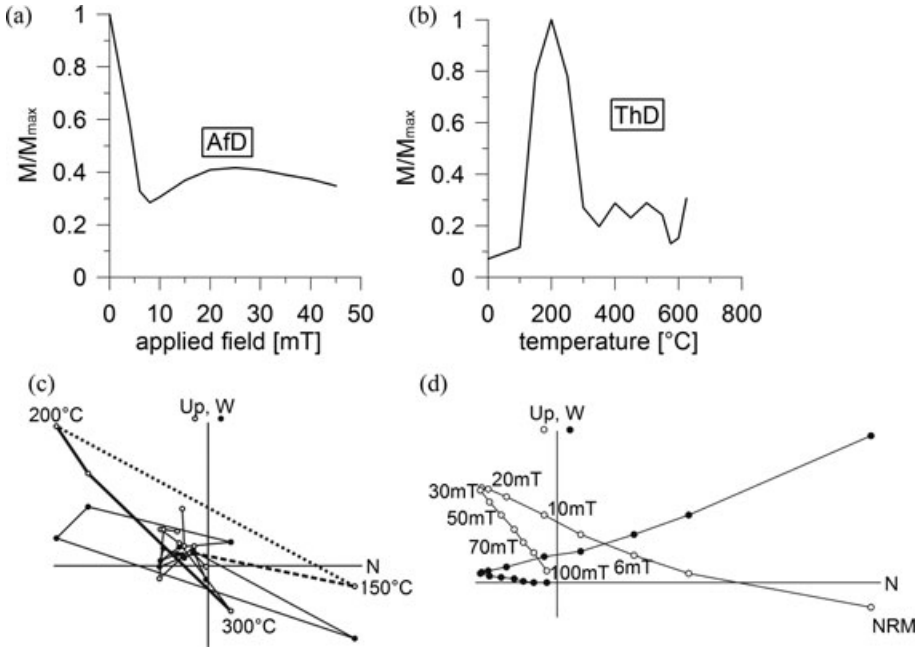



**Figure 11.** Combined alternating field demagnetization (up to 45 mT) and subsequent thermal demagnetization curve of LD 4–10–3. (a) Intensity curve of alternating field demagnetization up to 45 mT, (b) intensity curve of subsequent thermal demagnetization, (c) Zijderveld-diagram (geographic coordinates) of the thermal demagnetization; the thick full and dashed lines display different low-temperature components and (d) Zijderveld-diagram (geographic coordinates) of the twin specimen LD 4–10–2, which was demagnetized by alternating fields up to 100 mT.

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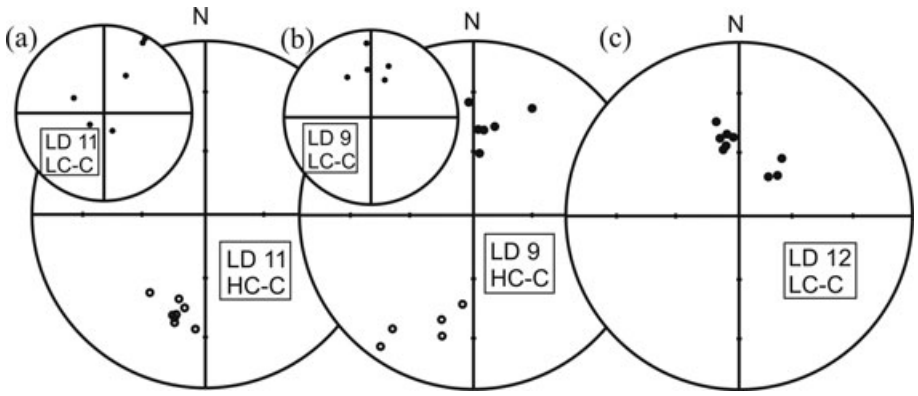



**Figure 12.** Equal area projection of *in situ* single specimen directions for representative sites: (a) LD 11, (b) LD 9 and (c) LD 12. Filled dots: positive inclination; open dots: negative inclination.

only two sites, it is rather ambiguous. Aitchison *et al.* (2007) also raised doubts about the reliability of Achache *et al.*'s data because of partial overprints. Nevertheless all presently available results are in good agreement except that one of Tan *et al.* The more northerly palaeolatitude derived by Tan *et al.* is mainly based upon 10 sites of volcanic tuffs of the T2 unit of the Linzizong Fm; it has to be asserted whether these tuffs were erupted during a short-term event.

Already existing Cretaceous data from red beds of the Takena Formation and intercalated lava flows show palaeolatitudes at the reference point of about 6.8–23°N for the Lhasa terrane (Fig. 15; Pozzi *et al.* 1982; Westphal *et al.* 1983; Achache *et al.* 1984; Lin & Watts 1988; Tan *et al.* 2010). Palaeolatitudes and their confidence limits of all referenced data are given in Table 2. These Cretaceous data post-date the closure of the Mesotethyan ocean and their age of deposition is dated to mid-Cretaceous (He *et al.* 2007; Leier *et al.* 2007). Lin & Watts (1988) additionally published Upper Cretaceous data of the Nagqu volcanics and the Qelico volcanics, both located on the northern Lhasa terrane. These data provide palaeolatitudes of 16.7° ± 4°N and 17.4° ± 11.1°N, respectively. All these results come from the Lhasa terrane at a longitude similar to the Linzhou Basin and the reference point used for calculating expected inclinations. One more Cretaceous palaeolatitude of the Lhasa terrane is available from the far western part (Chen *et al.* 1993), which fits to the other data in the central region. This indicates that the southern margin of the Lhasa terrane was rather straight before India–Asia collision. Conformity of Eocene and Cretaceous data indicates that no larger latitudinal motion of the Lhasa terrane occurred between Cretaceous and Early Eocene. This is in accordance with the APWP of Eurasia. The relatively constant discrepancies between observed and expected palaeolatitudes suggest that no considerable north–south directed crustal shortening occurred at the southern Eurasian margin between Cretaceous and Early Eocene. The dashed line (c) in Fig. 15 displays the pre-collisional southern margin of the Lhasa terrane according to our new result (13.2°N). The southern margin estimated from all available palaeomagnetic data is similar to this result (14.65°N). Regarding to the good quality of the palaeomagnetic data from the Linzhou Basin dykes and the consistency with previous results this can be considered as a reliable estimate.

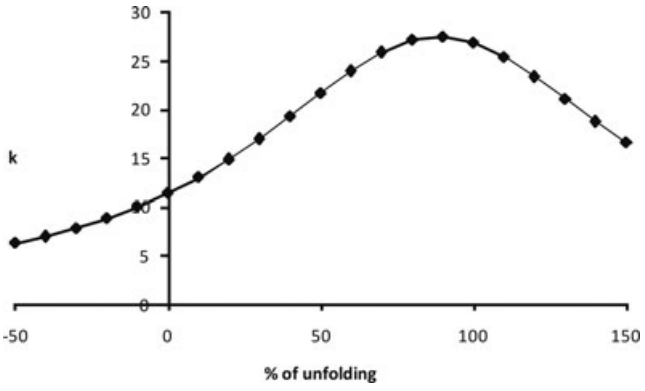



**Figure 13.** Results of stepwise unfolding of the mean direction of the HC-C from our dykes combined with nine previous palaeomagnetic sites of Achache *et al.* (1984) and Westphal *et al.* (1983).

### 6.2 Post-collisional tectonics and collision age

Similarities of expected and observed declinations indicate no block rotation of the Lhasa terrane since Early Eocene. Comparison of the present results with the APWP of Eurasia reveals a difference between expected and observed palaeolatitudes of 16.7° ± 6.9°. For determination of the confidence limit the Gaussian error propagation law was used including the 95 per cent confidence limits of the determined palaeolatitudes and of the virtual geomagnetic poles.

During the last two decades it was repeatedly observed that Late Cretaceous to Tertiary palaeomagnetic inclinations may have been anomalously low in central Asia (e.g. Westphal 1993; Thomas *et al.* 1994; Chauvin *et al.* 1996; Cogné *et al.* 1999) when compared to the APWP of Eurasia. Chauvin *et al.* (1996) showed that there is an increase in the inclination anomaly from 0° at the Atlantic margin to about 10° in central Europe and reaching a maximum of 25° in Central Asia. Several hypotheses were discussed concerning the origin of that anomaly (Cogné *et al.* 1999). Si & Van der Voo (2001) showed that the most likely explanations are (i) inclination shallowing due to depositional processes or compaction or (ii) a long-term non-dipole contribution to the geomagnetic field. In this study we drilled exclusively igneous rocks and therefore can exclude inclination shallowing due to syn-sedimentary or compaction-induced processes. Similar inclinations determined from the Early Cretaceous Takena Formation and the Eocene Linzizong Formation imply that if shallow inclinations exist due to a non-dipolar contribution to the geomagnetic field, this non-dipolar behaviour must have been stable throughout a very long period (>50 Myr). Previous observations about the inclination anomaly in central Asia mostly refer to Late Cretaceous to Tertiary data. Westphal (1993) and Chauvin *et al.*

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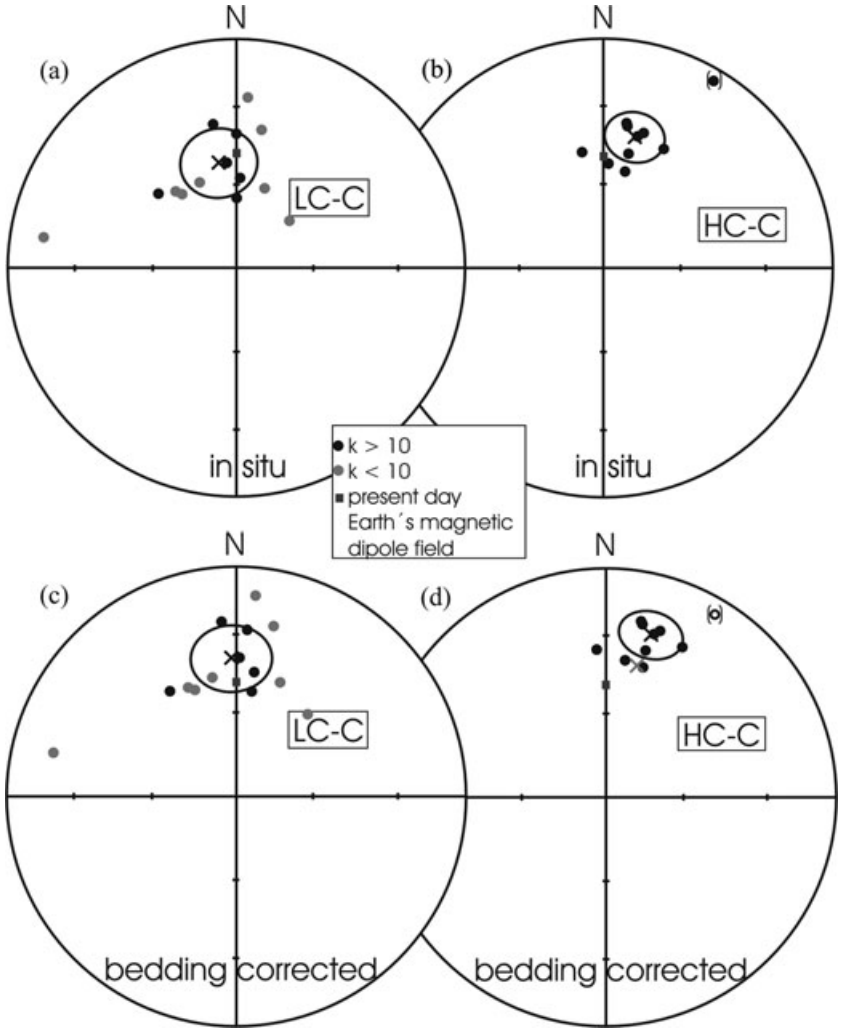



**Figure 14.** Equal area projection of site mean directions of (a) LC-C and (b) HC-C in geographic coordinates, and (c) LC-C and (d) HC-C in stratigraphic coordintes. The overall mean directions (black crosses), the 95 per cent confidence limits (circles) and the direction of the present day Earth magnetic dipole field (grey rectangles) are also shown. In case of the LC-C all site mean values are displayed, however, the overall mean direction and the 95 per cent confidence limit is based only upon sites with $k > 10$ (black filled circles). In case of the HC-C only sites with $k > 10$ are displayed.

(1996) proposed that a long-term anomaly was present during the Tertiary in central Asia. However, Gilder *et al.* (2003) presented palaeomagnetic data from Upper Jurassic to Lower Cretaceous volcanic and sedimentary rocks of the Tarim Basin, which fit well to the predicted palaeodirections and thus they exclude a long-standing non-dipole field for the region during that time. Furthermore Johnson & Constable (1997) and Carlut & Courtillot (1998) argue against long-lived non-dipolar, 'Pacific-Dipole-Window' sized anomalies. According to them poor site and/or temporal recording distorted earlier conclusions. The analysis of a worldwide database of 1536 Cenozoic and 1791 Mesozoic inclination observations by Kent & Smethurst (1998) yields that their frequency distributions are compatible with a time-averaged geomagnetic field which is very close to that of a geocentric axial dipole.

Therefore it is unlikely that the inclinations determined in our study are systematically influenced by a non-dipolar contribution to the geomagnetic field. The difference of 16.7° between expected and observed inclinations can be interpreted to represent a northward movement of the Lhasa terrane of 1847 ± 763 km from the time of India–Asia collision until present; expected palaeolatitudes of Eurasia remain stable during this period (Fig. 15). The northward movement of the Lhasa terrane is regarded as a result of the indentation of India into Eurasia and was predicted by several numerical models and laboratory experiments, which investigated the deformation of the Asian continent due to collision with India (e.g. England & McKenzie 1982; Tapponnier *et al.* 1982; Vilotte *et al.* 1982; England & Houseman 1985; Houseman & England 1986; Davy & Cobbold 1988; Houseman & England 1993; Cook & Royden 2008). The determined amount of northward movement reflects substantial north–south directed crustal shortening of the Tibetan Plateau accommodated by crustal thickening and lateral extrusion by strike-slip faulting and/or lower crustal flow (e.g. Royden 1996; Royden *et al.* 1997; Clark & Royden 2000; Beaumont *et al.* 2001; Tapponnier *et al.* 2001).

The age of initial collision between India and Eurasia can be determined by intersection of the pre-collisional southern margin of Eurasia and northern margin of India. It is generally accepted that the pre-collisional India was extended to the north ('Greater India'). The shape and size of Greater India is still a matter of debate. A review of different models has been published by Ali &

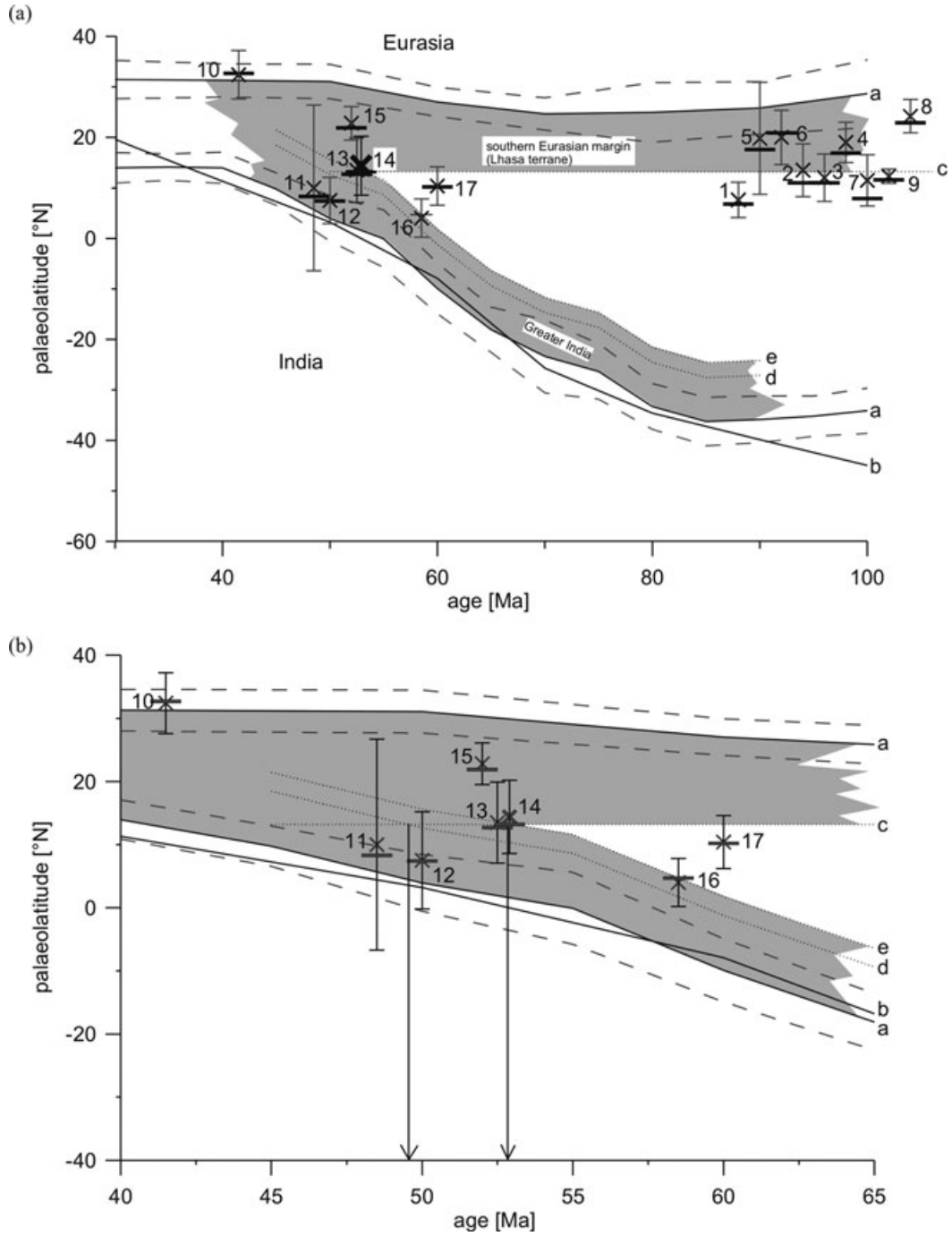



**Figure 15.** (a) Plot of palaeolatitudes of the Eurasian and Indian continents and the pre-collisional northern margin of India and southern margin of Eurasia. The solid lines (a) and (b) display expected palaeolatitudes calculated from APWPs (a: Besse & Courtillot 2002, b: Acton 1999) for a reference point at the Indus-Yarlung suture zone (29°N/90°E) based on the present day geographic configuration. Confidence limits for the APWPs of Besse & Courtillot (2002) are illustrated as dashed lines, the ones of Acton (1999) ($A_{95} = 4{,}7^{\circ}$) are not displayed. Crosses are palaeolatitudes calculated from palaeomagnetic studies on the Lhasa terrane itself, the error bars show d$p$ values; horizontal bars display the palaeolatitues after correction for the distance between sampling location and the reference point. The dashed line (c) corresponds to the southern Eurasian margin derived from our results of the Linzhou basin dykes (13.2°N). The extended Eurasian margin estimated by previous and present palaeomagnetic studies comprises almost the same value (14.65°N) and is not shown. The dashed lines (d) and (e) display the pre-collisional northern extent of Greater India (d: Ali & Aitchison 2005, e: Patzelt *et al.* 1996—for further explanation see text). References of palaeomagnetic data: (1) Lin & Watts 1988 (Takena Fm), (2) Lin & Watts 1988 (Qelico volcanics), (3) Achache *et al.* 1984 (Takena Fm southern part), (4) Achache *et al.* 1984 (Takena Fm northern part), (5) Lin & Watts 1988 (Nagqu volcanic), (6) Westphal *et al.* 1983 (Takena Fm), (7) Chen *et al.* 1993, (8) Tan *et al.* 2010 (lava flows), (9) Tan *et al.* 2010 (red beds of Takena Fm), (10) Tan *et al.* 2010 (Linzizong Fm), (11) Westphal *et al.* 1983 (Linzizong Fm), (12) Chen *et al.* 2010 (T3 & T2 unit of Linzizong Fm), (13) Achache *et al.* 1984 (Linzizong Fm), (14) this paper (dykes Linzhou Basin), (15) Dupont-Nivot *et al.* 2010 (K/T & T1 unit of Linzizong Fm), (16) Patzelt *et al.* 1996 (Zongpu Fm) and (17): Chen *et al.* 2010 (K/T & T1 unit of Linzizong Fm). For better visualization of confidence limits the results from the Taken Fm are plotted with a 2-Myr relative shift against each other. (b) Same plot zoomed on a time interval between 40 and 60 Ma. Lines and symbols are the same as explained in plot (a). Intersection of the southern Eurasian margin and the northern Indian margin and the correlated India–Asia collision times are marked by arrows.

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**Table 2.** Previous palaeomagnetic data from the Lhasa terrane.

| Location | Lat. | Long. | Age | *N* | Pallat | Pallat corr | d*p* | Ref. |
|---|---|---|---|---|---|---|---|---|
| Linzizong Fm (T3 unit) | 30,0 | 91,2 | lEc | 10 | 32,4 | 32,7 | 4,8 | Tan *et al.* (2010) |
| Linzizong Fm (T3 & T2 unit) | 29,9 | 90,2 | lEc | 15 | 7,5 | 7,4 | 4,6 | Chen *et al.* (2010) |
| Linzizong Fm (K/T & T1 unit | 29,9 | 90,2 | lEc | 23 | 10,4 | 10,2 | 3,8 | Chen *et al.* (2010) |
| Linzizing Fm | 30,0 | 91,1 | lEc | 37 | 22,8 | 21,9 | 3,3 | Dupont-Nivet *et al.* (2010) |
| Linzizong Fm | 29,9 | 91,0 | lEc | 8 | 13,5 | 12,7 | 6,4 | Achache *et al.* (1984) |
| Linzizong Fm | 29,9 | 91,0 | lEc | 2 | 10,0 | 8,3 | 16,7 | Westphal *et al.* (1983) |
| Takena Fm (red beds) | 29,9 | 91,2 | uKr | 43 | 12,3 | 11,6 | 1,4 | Tan *et al.* (2010) |
| Takena Fm (lava flows) | 29,9 | 91,2 | uKr | 21 | 24,2 | 22,9 | 3,3 | Tan *et al.* (2010) |
| Takena Fm north | 32,0 | 91,5 | uKr | 6 | 13,5 | 11,0 | 5,2 | Achache *et al.* (1984) |
| Takena Fm south | 29,9 | 91,0 | uKr | 8 | 12,0 | 11,0 | 4,7 | Achache *et al.* (1984) |
| Takena Fm | 29,9 | 91,2 | uKr | 8 | 7,6 | 6,8 | 3,5 | Lin & Watts (1988) |
| Takena Fm | 29,9 | 91,0 | uKr | 6 | 20,0 | 20,9 | 5,4 | Westphal *et al.* (1983) |
| Qelico volc. | 32,7 | 91,0 | uKr | 4 | 19,8 | 17,6 | 11,1 | Lin & Watts (1988) |
| Nagqu volc. | 31,5 | 92,0 | uKr | 9 | 19,0 | 16,9 | 4,0 | Lin & Watts (1988) |
| Domar | 33,7 | 80,4 | Kr | 14 | 11,5 | 7,9 | 5,1 | Chen *et al.* (1993) |
| Zongpu Fm | 28,2 | 88,9 | Pc | 14 | 4,0 | 4,7 | 3,8 | Patzelt *et al.* (1996) |

Lat.: latitude; Long.: longitude; *N*: number of sites; pallat: palaeolatitude; pallat corr: palaeolatitude for a reference point on the suture zone (20°N/90°E); Ref: reference; lEC: lower Eocene; (u)Kr: (upper) Cretaceous; Pc: Palaeocene.

Aitchison (2005). The proposed models show a large variation of the north–south extent ranging from a few kilometres (Gnos *et al.* 1997) to over 2800 km (Rotstein *et al.* 2001). Most models are based on indirect approaches, for example, reconstruction of Gondwana, estimates of shortening between the Indian craton and the IYSZ and determination of the gap between India and Eurasia at the expected collision time. Based on bathymetric features in the eastern Indian Ocean Ali & Aitchison (2005) proposed a Greater India with 950 km extension in the central part and about 500 and 600 km in the east and west. They point out that there are absolute limits for the extent of Greater India defined by the possible refit into Gondwana and bathymetric features in the eastern Indian Ocean. Therefore, models of Greater India considerably bigger or smaller than the one of Ali & Aitchison (2005) were considered as rather unlikely. Using palaeomagntic investigations on mid-Cretaceous to Palaeocene sediments in the Tethyan Himalayas at Gamba and Duela in southern Tibet, Patzelt *et al.* (1996) determined the palaeolatititude of the northern margin of India. Because of the good recorder properties (positive fold test, stable high coercive magnetite remanence) we used here only their result for the (Upper Palaeocene) Zongpu Formation, which yield a palaeolatitude of 4°N. Both models of Greater India, the one proposed by Ali & Aitchison (2005) and the one proposed by Patzelt *et al.* (1996) have similar extensions and both seem to be reliable. The result of Ali & Aitchison mentions an extent and the position of the northern margin of India [dashed line (d) in Fig. 15] has to be determined by a relative distance to the expected position obtained from the APWP of Besse & Courtillot (2002). The result of Patzelt *et al.* provides a direct palaeolatitude for the northern margin [dashed line (e) in Fig. 15]. For comparison between direct and indirect approaches we used both results to determine the collision age. Intersection of the lines for the southern Eurasian margin and the northern Indian margin defines the time of the initial contact of the Eurasian and Indian continents, that is, 53–49 ± 6 Ma. Error estimates are based on the 95 per cent confidence limits for the palaeolatitudes of the northern Indian margin (Patzelt *et al.* 1996) and the southern Eurasian margin (this study). An initial collision within this time range is in accordance with previous studies based on palaeomagnetic or geologic data (e.g. Patriat & Achache 1984; Klootwijk *et al.* 1992; Critelli & Garzanti 1994; Privnik & Wells 1996; Guillot *et al.* 2003). Our result for the collision age is a minimum estimate and could be even older if the former Greater India was extended significantly further to the north of Gamba and Duela areas. Although there is no geological record of such an extension, a larger northern continental margin of Greater India could have been underthrusted or subducted beneath the Lhasa terrane along the IYSZ. Thus an earlier collision at 60–65 Ma as suggested by Ding *et al.* (2005) may be still possible. In either case, a significantly later collision age at the Eocene/Oligocene boundary (34 Ma) as recently proposed by Aitchison *et al.* (2007) is not in agreement with our new and previous palaeomagnetic data from the Lhasa block.

## 7 CONCLUSIONS

Remanence directions residing in a LC-C and HC-C were separated. Ti-rich titanomagnetite was revealed as the remanence carrier of the HC-C, which is indicative for a fresh rock and ascertains the reliability of a primary remanence. This is supported by a positive fold test.

An Early Eocene palaeolatitude of 13.2° ± 5.8°N was determined for the Lhasa terrane at a reference point on the YZSZ (29°N/90°E). The differences of expected and observed palaeolatitudes yields a northward movement of the Lhasa terrane of 1847 ± 763 km since Early Eocene resulted by the indentation of India into Asia. It confirms a large north–south crustal shortening between Early Eocene and present. Comparison of expected and observed declinations yields no tectonic rotation of the central Lhasa terrane since Early Eocene.

Comparison of Eocene and Creatceous palaeomagnetic data of the Lhasa terrane shows no significant latitudinal motion of the Lhasa terrane from mid-Creatceous to Early Eocene, which is in accordance with the APWP of Eurasia; constant difference between expected and observed palaeolatitudes during this period indicates that no considerable north–south crustal shortening within the southern Eurasian margin occurred within that time.

A collision age between c. 53–49 (±6) Ma was determined using the results for the southern margin of Eurasia according to our new data and all available ones as well as models for 'Greater India' of Ali & Aitchison (2005) and Patzelt *et al.* (1996). A very late collision age as assumed by Aitchison *et al.* (2007) is clearly disproved.

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